Lectures31_MA5104

14 October 2026 11:29

Lectures31
_MA5104

MA 5104: Statistical Inference and Learning

Module 4: Learning
Lecture 31: Logistic Regression and GLMs

Dr. Rakhi Singh
Department of Mathematics
IIT Madras

Fall 2026

Regression versus Classification

Everything so far assumed the response Y was quantitative.

In many situations it is qualitative: whether a customer is risky, whether a patient has a disease, which of several products a buyer chooses.

Predicting a qualitative response is called classifying the observation — assigning it to a category.

The shift in target

Regression estimates $\mathbb{E}[Y \mid X]$.
Classification estimates $P(Y = k \mid X)$, and the Bayes rule $G^*(x) = \arg\max_k P(Y = k \mid X = x)$.

Many methods do this: logistic regression, Poisson regression, LDA, QDA, naive Bayes, kNN, classification trees. Today: the first two, and the framework they belong to.

Different Situations

- A person arrives at the emergency room with symptoms attributable to one of three conditions. Which one?
- An online banking service must decide whether a transaction is fraudulent, from the IP address, transaction history, and so on.
- From DNA sequence data on patients with and without a disease, a biologist wants to identify which mutations are disease-causing.

An example

Predict whether an individual will `default` on a credit card payment, from annual `income` and monthly `balance`.

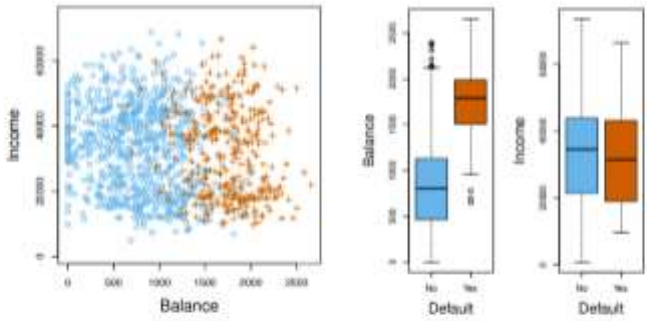


FIGURE 4.1. *The* `Default` *data set.* Left: *The annual incomes and monthly credit card balances of a number of individuals. The individuals who defaulted on their credit card payments are shown in orange, and those who did not are shown in blue.* Center: *Boxplots of* `balance` *as a function of* `default` *status.* Right: *Boxplots of* `income` *as a function of* `default` *status.*

Why Not Just Code the Categories?

Predict eye colour from genes. Code

$$Y = 1 \text{ (brown)}, \quad 2 \text{ (black)}, \quad 3 \text{ (blue)}$$

A positive coefficient would mean "higher x moves you from black towards blue". That is not a sentence about anything.

Now predict the grade in this course, where the categories *are* ordered:

$$Y = 1 \text{ (S)}, \quad 2 \text{ (A)}, \quad 3 \text{ (B)}, \quad 4 \text{ (C)}$$

The ordering is real, but least squares additionally assumes the gap between S and A equals the gap between B and C.

Coding categories as numbers smuggles in an ordering, and a spacing. Neither is usually justified.

A Second Look: Heart Disease

Three variables: `chd` (heart disease indicator), `height`, and `cigs` (cigarettes per day).

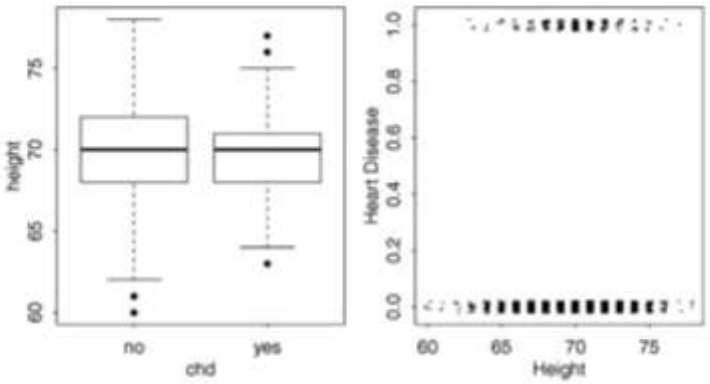


Figure 2.1 *Plots of the presence/absence of heart disease according to height in inches.*

A Second Look: Heart Disease

Three variables: `chd` (heart disease indicator), `height`, and `cigs` (cigarettes per day).

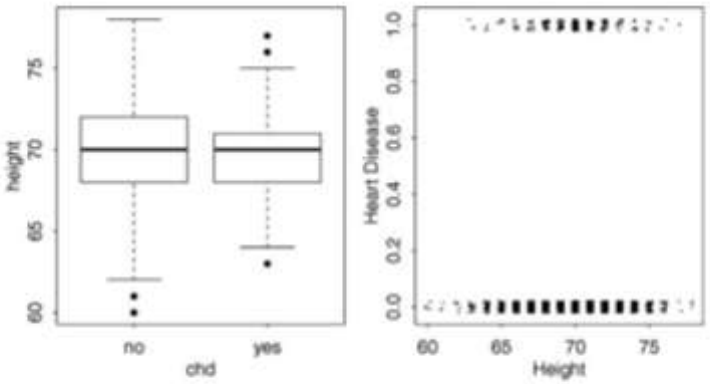


Figure 2.1 *Plots of the presence/absence of heart disease according to height in inches.*

Heights look similarly distributed in both groups.

A Second Look: Heart Disease

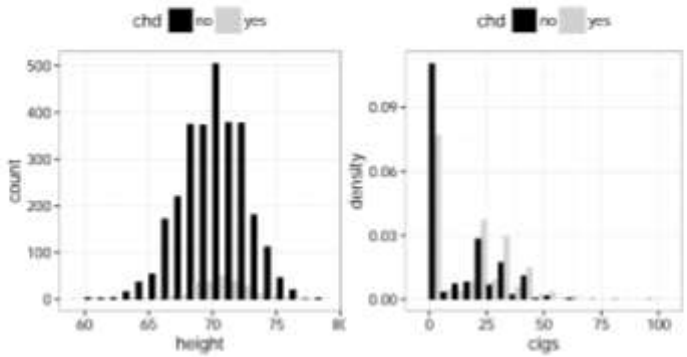


Figure 2.2 *Interleaved histograms of the distribution of heights and cigarette usage for men with and without heart disease.*

A Second Look: Heart Disease

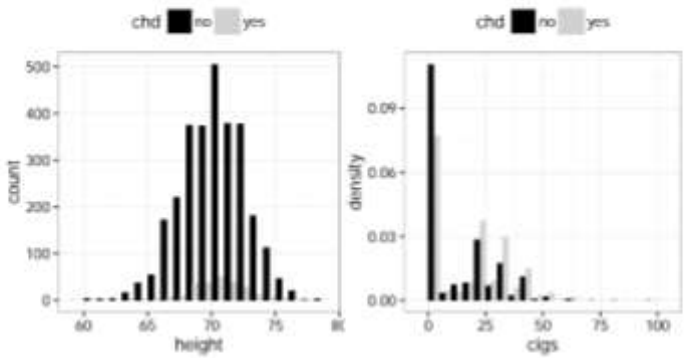


Figure 2.2 *Interleaved histograms of the distribution of heights and cigarette usage for men with and without heart disease.*

Smokers appear more likely to have heart disease.

A Second Look: Heart Disease

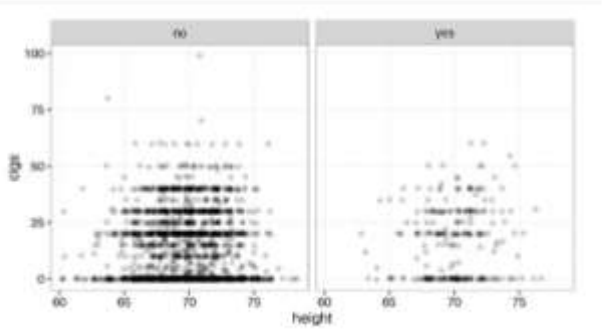


Figure 2.3 *Height and cigarette consumption for men without heart disease on the left and with heart disease on the right. Some jittering and transparency have been used to reduce overplotting problems.*

The problem: at the *same* height and cigarette consumption, both 0 and 1 occur routinely. So we should not try to model the outcome. We should model the probability of the outcome.

Why a Linear Model on Probabilities Fails

Suppose we fit $p(x) = \beta_0 + \beta_1 x$ directly.

Why a Linear Model on Probabilities Fails

Suppose we fit $p(x) = \beta_0 + \beta_1 x$ directly.

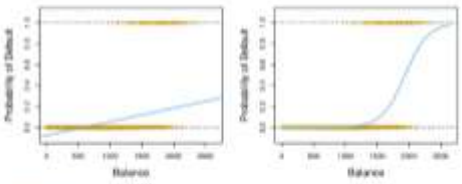


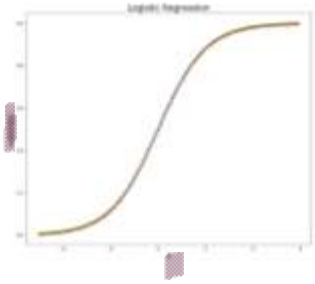
FIGURE 4.2. *Classification using the* `Default` *data. Left: Estimated probability of* `default` *using linear regression. Some estimated probabilities are negative! The orange ticks indicate the 0/1 values coded for* `default` *(No or Yes). Right: Predicted probabilities of* `default` *using logistic regression. All probabilities lie between 0 and 1.*

- A line eventually goes outside [0,1]. Not probabilities.
- We could truncate to $[0, 1]$, but then there are regions of predictor space where the outcome is declared certain. Not reasonable here.

We need a function that is linear in the parameters but whose output is confined to $(0, 1)$.

The Logistic Model

$$P(Y=1 \mid X=x) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}} = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}}$$



- For low balances the predicted probability is close to, but never below, zero; for high balances close to, but never above, one.
- Always an S-shaped curve, so *every* value of x gives a sensible prediction.
- Parameters are estimated by maximum likelihood.

Odds and Log-Odds

Rearranging the model,

$$\frac{P(Y=1 \mid X)}{1-P(Y=1 \mid X)} = e^{\beta_0+\beta_1 x}, \quad \text{so} \quad \log\left(\frac{P(Y=1 \mid X)}{P(Y=0 \mid X)}\right) = \beta_0+\beta_1 x$$

The left-hand side is the log-odds, or logit. The logistic model is the model whose logit is linear in x.

Reading the numbers. If $P(Y=1 \mid X)=0.2$, the odds are $1/4$: on average 1 person in 5 defaults.

The contrast to hold on to

In linear regression, $\hat{\beta}_1$ is the average change in Y per unit increase in x.
In logistic regression, $\hat{\beta}_1$ is the change in the log-odds per unit increase in x and $e^{\hat{\beta}_1}$ is an odds ratio.

The Link Function, and Its Alternatives

The linear link $p_i = \beta_0 + \beta_1 z_i$ cannot work: it does not keep $p_i \in [0,1]$.
The logistic link does:

$$\text{logit}(p_i) = \log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 z_i = L \qquad \Longrightarrow \qquad p_i = \frac{1}{1+e^{-L}} \in (0,1)$$

- $L \in (-\infty, \infty)$ maps onto $p \in (0,1)$; $L = 0$ gives $p = 0.5$; the function is symmetric in L.
- $\beta_1 > 0$: $p(z)$ increasing. $\beta_1 < 0$: decreasing. $\beta_1 = 0$: p does not depend on z.

Other links also work. The probit link uses the standard normal CDF:

$$\Phi^{-1}(p_i) = \beta_0 + \beta_1 z_i \qquad \Longrightarrow \qquad p_i = \Phi(\beta_0 + \beta_1 z_i)$$

Where the Likelihood Comes From

Recall a binomial experiment: n independent trials, each $Y_i \in \{0,1\}$. One requirement changes: the success probability is allowed to differ from trial to trial, and to depend on covariates,

$$p_i(z_i) = P(Y_i = 1 \mid Z_i = z_i)$$

With $P(Y_i = x) = p_i^x (1-p_i)^{1-x}$, the likelihood is

$$L_n(p_1, \ldots, p_n) = \prod_{i=1}^{n} p_i^{x_i} (1-p_i)^{1-x_i}$$

Why we must impose structure

If all the p_i are free, the MLE is $\hat{p}_i = x_i$, that is, each probability is 0 or 1. Useless. The link function is what ties the p_i together through a handful of parameters.

The Log-Likelihood

Substituting the logistic link,

$$L_n(\beta_0, \beta_1) = \prod_{i=1}^{n} \left(\frac{e^{\beta_0+\beta_1 z_i}}{1+e^{\beta_0+\beta_1 z_i}} \right)^{x_i} \left(\frac{1}{1+e^{\beta_0+\beta_1 z_i}} \right)^{1-x_i}$$

and taking logs,

The Log-Likelihood

Substituting the logistic link,

$$L_n(\beta_0, \beta_1) = \prod_{i=1}^{n} \left(\frac{e^{\beta_0+\beta_1 z_i}}{1+e^{\beta_0+\beta_1 z_i}} \right)^{x_i} \left(\frac{1}{1+e^{\beta_0+\beta_1 z_i}} \right)^{1-x_i}$$

and taking logs,

$$\boxed{\ell_n(\beta_0, \beta_1) = \sum_{i=1}^{n} x_i(\beta_0 + \beta_1 z_i) - \sum_{i=1}^{n} \log\left(1 + e^{\beta_0+\beta_1 z_i}\right)}$$

- This is concave, so there is a unique maximum.
- There is no closed form. The MLE is obtained numerically.
- For the `Default` data,
  - the p-value for `balance` is tiny, so we reject $H_0 : \beta_1 = 0$.
  - The intercept is rarely of interest. It exists to make the average fitted probability match the observed proportion of ones.

Fitting: Newton–Raphson

Second-order Taylor expansion of the log-likelihood about the current value:

$$\ell(\beta) \approx \ell(\beta^{\text{old}}) + \nabla\ell(\beta^{\text{old}})^{\mathsf{T}}(\beta - \beta^{\text{old}}) + \tfrac{1}{2}(\beta - \beta^{\text{old}})^{\mathsf{T}} \nabla^2\ell(\beta^{\text{old}})(\beta - \beta^{\text{old}})$$

Set the gradient of the approximation to zero:

$$\nabla\ell(\beta^{\text{old}}) + \nabla^2\ell(\beta^{\text{old}})(\beta - \beta^{\text{old}}) = 0$$

giving the update

$$\boxed{\beta^{\text{new}} = \beta^{\text{old}} - \left(\nabla^2\ell(\beta^{\text{old}})\right)^{-1} \nabla\ell(\beta^{\text{old}})}$$

Exactly the method we used on least squares. There the Hessian was constant, so one step landed on the answer. Here it depends on β, so we must iterate.

Regularized Logistic Regression

Add a penalty to the negative log-likelihood:

$$\min_w \; -\ell(w) + \lambda \|w\|_2^2 \qquad \text{or} \qquad \min_w \; -\ell(w) + \lambda \|w\|_1$$

- Controls complexity: penalizes large coefficients and prevents overfitting in high-dimensional or noisy settings.
- ℓ_2 shrinks coefficients smoothly and improves numerical stability; ℓ_1 promotes sparsity.
- This is ridge and lasso, for classification. Nothing new — the same penalty on a different loss.
- λ trades data fit against complexity, and is chosen the same way as before.

A practical note

Regularization also rescues the fit under perfect separation, where the unpenalized MLE diverges.

The Confusion Matrix, and Why Accuracy Misleads

Same fit, threshold 0.5:

	predicted default	predicted no
actually default	55	114
actually no	28	803

Accuracy $= (55 + 803)/1000 = 0.858$. A good model?

The Confusion Matrix, and Why Accuracy Misleads

Same fit, threshold 0.5:

	predicted default	predicted no
actually default	55	114
actually no	28	803

Accuracy $= (55 + 803)/1000 = 0.858$. A good model?

Predict "no default" for *everyone* and accuracy is 0.831. The model buys three percentage points — while missing two thirds of the defaulters (sensitivity 0.325).

Sensitivity (recall) $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$ Specificity $= \mathrm{TN}/(\mathrm{TN} + \mathrm{FP})$ Precision $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$

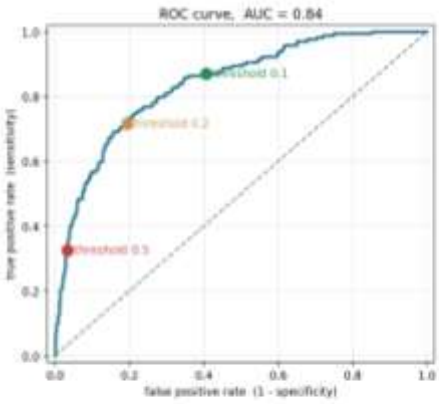
The Threshold Is a Separate Decision

threshold	TP	FP	FN	TN	accuracy	sensitivity	specificity
0.5	55	28	114	803	0.858	0.325	0.966
0.2	121	162	48	669	0.790	0.716	0.805
0.1	147	338	22	493	0.640	0.870	0.593

Identical model, identical fitted probabilities. Only the cutoff moved.

Lowering the threshold catches more defaulters and raises more false alarms. Which trade you want depends on the relative cost of the two errors — and 0.5 is the right answer *only* when they cost the same.

ROC: The Whole Menu of Thresholds



Sweep the threshold from 1 to 0 and plot sensitivity against 1−specificity. The three points are the three rows of the previous table.

- The diagonal is random guessing; the top-left corner is perfection.
- AUC = 0.84: Prob. that a randomly chosen defaulter is scored above a randomly chosen non-defaulter.
- AUC is threshold-free, which is its strength and its weakness.

Multiple Predictors, and a Reversal

	Coefficient	Std. error	z-statistic	p-value
Intercept	−3.5041	0.0707	−49.55	<0.0001
student[Yes]	0.4049	0.1150	3.52	0.0004

TABLE 4.2. *For the* Default *data, estimated coefficients of the logistic regression model that predicts the probability of* default *using student status. Student status is encoded as a dummy variable, with a value of 1 for a student and a value of 0 for a non-student, and represented by the variable* student[Yes] *in the table.*

	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.8690	0.4923	−22.08	<0.0001
balance	0.0057	0.0002	24.74	<0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	−0.6468	0.2362	−2.74	0.0062

Qualitative predictors enter through dummy variables, exactly as in linear regression.

And, exactly as in linear regression, results from one predictor can differ sharply from results with several — especially when predictors are correlated.

Students tend to carry large balances, and large balances mean higher default rates.

The distinction that matters to a lender

A student is riskier than a non-student if you know nothing about the balance. But that student is less risky than a non-student *with the same balance.*

Multicategory Logit: The Naive Attempt

Let Y be nominal with J categories and response probabilities $p = (p_1, \ldots, p_J)$, $\sum_j p_j = 1$. With n independent observations the counts are Multinomial(n, p).

Take $J = 3$. Why not fit three separate binary logits?

$$\log \frac{p_{i,1}}{1-p_{i,1}} = \alpha_1 + \beta_1 x_i, \quad \log \frac{p_{i,2}}{1-p_{i,2}} = \alpha_2 + \beta_2 x_i, \quad \log \frac{p_{i,3}}{1-p_{i,3}} = \alpha_3 + \beta_3 x_i$$

This does not work: the fitted probabilities are not constrained to satisfy $p_{i,1} + p_{i,2} + p_{i,3} = 1$ for every i.

So what should we do instead?

Baseline-Category Logits

Model each *pair* of categories instead:

$$\log \frac{p_{i,1}}{p_{i,2}}, \qquad \log \frac{p_{i,1}}{p_{i,3}}, \qquad \log \frac{p_{i,2}}{p_{i,3}}$$

These are mutually consistent, and one is redundant:

$$\log \frac{p_{i,2}}{p_{i,3}} = \log \frac{p_{i,1}}{p_{i,3}} - \log \frac{p_{i,1}}{p_{i,2}}$$

With J categories there are $\binom{J}{2}$ pairs but only $J-1$ need to be specified. Choosing the Jth as baseline:

$$\log \frac{p_{i,j}}{p_{i,J}} = \alpha_j + \beta_j x_i, \qquad j = 1, \ldots, J-1$$

$$p_{i,j} = \frac{e^{\alpha_j + \beta_j x_i}}{1 + \sum_{m=1}^{J-1} e^{\alpha_m + \beta_m x_i}}, \qquad p_{i,J} = \frac{1}{1 + \sum_{m=1}^{J-1} e^{\alpha_m + \beta_m x_i}}$$

The choice of baseline is arbitrary and is made for interpretability.

What Changes When $J > 2$

- The $J-1$ logit equations are estimated simultaneously. The fitted probabilities are then the same whichever category is chosen as baseline.
- The number of parameters is $J-1$ times that of a binary logistic model. Every predictor buys $J-1$ coefficients.
- Consequently the sample size must be large enough to support all those functions. If it is not, you will see estimation failures and warnings about infinite estimates. Reducing the number of response categories can help.

Interpretation changes too

The coefficients (α_j, β_j) are effects on contrasts between pairs of categories — never on the probability of being in a particular category. In the binary case a positive coefficient means a higher probability of the outcome. In the multinomial model that is not always true.

When the Categories Are Ordered

Ignoring the ordering and fitting generalized logits is *not wrong*, but models that use the ordering have simpler interpretations and often greater power.

The cumulative logit model uses $P(Y \le j) = p_1 + \cdots + p_j$:

$$P(Y \le j) = \log \frac{p_1 + \cdots + p_j}{p_{j+1} + \cdots + p_J} = \alpha_j + \beta_j x, \qquad j = 1, \ldots, J-1$$

Each cumulative logit collapses Y into two categories — $\{1, \ldots, j\}$ against $\{j+1, \ldots, J\}$ — and uses all J categories.

- Other options: adjacent-categories logits, continuation-ratio logits. For $J = 2$ all of these reduce to ordinary binary logit.
- Caution: ordinal models impose restrictions that may be inappropriate, and the fitted probabilities are not guaranteed positive. Check the restrictions.

Count Responses: Poisson Regression

Suppose Y is a *count* — website visitors, accidents per week. Least squares has three problems: predictions can be negative, the variance is not constant, and the fitted values must be rounded.

Instead let $Y_i \sim \text{Poisson}(\mu_i)$ and use the log link:

$$\log(\mu_i) = x_i^\top \beta \qquad \Longrightarrow \qquad \mu_i = \exp(x_i^\top \beta) > 0$$

- **Interpretation:** a one-unit increase in x_j multiplies $\mathbb{E}(Y)$ by e^{β_j}. In the bike-share data, cloudy rather than clear skies gives $e^{-0.08} = 0.923$: on average only 92.3% as many riders.
- **Mean–variance:** the Poisson model asserts mean = variance, which varies with i. Linear regression asserts constant variance.
- **Nonnegativity:** no negative predictions, unlike linear regression, which gave negative predictions for almost 10% of the bike-share data.

Generalized Linear Models

We have now seen three cases: Gaussian, binomial, Poisson. They share a structure. A GLM is specified by two things.

1. A response from the exponential family:

$$f(y \mid \theta, \phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right]$$

θ is the canonical parameter (location), ϕ the dispersion (scale). For the Gaussian ϕ is free; for Poisson and binomial it is fixed at 1, because those are one-parameter families.

The mean depends on θ alone; the variance factorises as $\text{Var}(\mu) = b''(\theta)a(\phi)$.

Link function and fitting Any GLM

2. A link function relating the mean to the linear predictor:

$$\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = x^\top \beta, \qquad \eta = g(\mu)$$

The canonical link is the g with $g(\mu) = \theta$. It is convenient and usually natural, but it is not compulsory.

The Newton Raphson idea generalizes. If we could compute the working responses Z_i we would fit the model by weighted least squares of Z on X, with weights $1/V(Z_i)$. Since those quantities are unknown, we iterate — and

$$\widehat{\mathrm{Var}}(\hat{\beta}) = (X^\top W X)^{-1} \hat{\phi}$$

with W the diagonal matrix of weights.

Four Things to Take Away

1. Model the probability, not the outcome. At the same predictor values both 0 and 1 occur, and a line on the probability scale leaves $[0, 1]$.
2. The logit is linear; the probability is not. $\hat{\beta}_j$ is a change in log-odds, so $e^{\hat{\beta}_j}$ is an odds ratio, and the effect on the probability depends on where you are on the curve.
3. Fitting is Newton–Raphson, and each step is a weighted least squares. The likelihood is concave with a unique maximum, but there is no closed form because the Hessian depends on β.
4. With J categories every predictor costs $J - 1$ parameters, and the coefficients describe contrasts against a baseline, not the probability of any single category.